

David Bose

Embedded Systems & IoT Firmware Engineer • 12 Years Experience • MERIQ Index: 73% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, I2C/SPI/UART, ARM Cortex, PCB Debugging, Python, MQTT, FreeRTOS

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — Intel (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 30%.
- Led a squad of 4 engineers delivering scalable microservices processing 33M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 32%.

Associate Embedded Systems & IoT Firmware Engineer — Bosch (2014 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using ARM Cortex saving \$22k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [C/C++, I2C/SPI/UART, ARM Cortex, PCB Debugging]

Engineered end-to-end production solution leveraging C/C++ and I2C/SPI/UART with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [ARM Cortex, PCB Debugging, Python, MQTT]

Architected high-resilience system integrating ARM Cortex with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

IIIT Hyderabad — 2010 - 2014 (Score: 3.83 / 4.0)